

CUHK(SZ)-CSC2003 Mid-term Exam

Mar. 20, 2024

Name:

Student ID:

Note:

- a. This is a closed-book exam.
- b. This exam paper has six pages in single-sided printing, including the cover page.
- c. Answer all questions within 90 minutes.
- d. The student must NOT take the exam paper away from the exam venue.
- e. Questions are not ordered by difficulty.

1. Circle **ONE** solution that best suits each question.

(a) (4 points) What is the extension of java source code files?

- A. .js
- B. .txt
- C. .class
- D. .java

(b) (4 points) Which of the following Java statements declares and allocates a 2-dimensional array of integers with six columns and four rows

- A. `int array[6][4];`
- B. `int array[4][6];`
- C. `int[ ][ ] array = new int[6][4];`
- D. `int[ ][ ] array = new int[4][6]`

(c) (4 points) Which of the following best defines a data type?

- A. A set of values.
- B. A set of operations.
- C. A set of values and operations on those values.
- D. A sequence of 0s and 1s.

(d) (4 points) How many bytes does an `int` variable occupy in Java?

- A. 1 byte
- B. 2 bytes
- C. 4 bytes
- D. 8 bytes

(e) (4 points) What does the following code snippet print?

```
1  for(int i = 0; i < 5; i+=2) {  
2      System.out.print(i + " ");  
3  }
```

- A. 0 2 4
- B. 0 1 2 3 4
- C. 1 2 3 4
- D. 0 1 2 3 4 5

2. (18 points) Give the value and type of each of the following Java expressions. For any expression that will not compile, write **X** for its value and **Illegal** for its type.

Expression	Value	Type
<code>Integer.parseInt("123")</code>		
<code>1 + 23 + "45"</code>		
<code>1 = 2</code>		
<code>(1 < 2) == false</code>		
<code>3 * 0.5 + 3 / 2 * 0.5</code>		
<code>(double)1/4</code>		

3. Consider the following Java program, and suppose `StdIn` and `StdOut` can be used smoothly:

```

1 public class Q3
2 {
3     public static void main(String[] args)
4     {
5         int x = StdIn.readInt();
6         int y = Integer.parseInt(args[1]);
7         StdOut.print(x + y + " ");
8         StdOut.print(x + " ");
9         StdOut.println(y);
10    }
11 }

```

The file `input.txt` contains the following text:

```

1 10 9 8 7

```

- (a) (6 points) Write the result of executing the following command.

```

1 java Q3 3 4 < input.txt

```

- (b) (6 points) Write the result of executing the following command.

```
1 java Q3 3 4 < input.txt | java Q3 2 3
```

4. (16 points) Consider the following program.

```
1 public class Q4
2 {
3     public static void f(int[] a, int m)
4     {
5         double[] b = f(a, 6.0);
6         a[0] = a[0] * m;
7         a[2] = (int) (b[0] + b[1]);
8         m = m + 5;
9     }
10    public static double[] f(int[] a, double x)
11    {
12        double[] b = new double[2];
13        b[0] = a[0] + x;
14        b[1] = a[1] + x;
15        return b;
16    }
17    public static void main(String[] args)
18    {
19        int[] a = {2, 3, 4};
20        int k = 5;
21        f(a, k);
22        StdOut.println(k);
23        StdOut.println(a[0]);
24        StdOut.println(a[1]);
25        StdOut.println(a[2]);
26    }
27 }
```

Give the output of this program in the blanks provided below.

5. (15 points) Consider the following program,

```
1 public class Q5
2 {
3     public static String convert(int N)
4     {
5         if (N == 1) return "1";
6         return convert(N/2) + (N % 2);
7     }
8     public static void main(String[] args)
9     {
10        int N1 = Integer.parseInt(args[0]);
11        int N2 = Integer.parseInt(args[1]);
12        int N3 = Integer.parseInt(args[2]);
13        StdOut.println(convert(N1));
14        StdOut.println(convert(N2));
15        StdOut.println(convert(N3));
16    }
17 }
```

Give the output of this program when running `java Q5 1 2 10`

6. (10 points) Consider the following Java program,

```
1 public class Q6 {
2     public static void main(String[] args) {
3         int[] a = new int[args.length];
4         for (int i = 0; i < args.length; i++)
5             a[i] = Integer.parseInt(args[i]);
6         for (int i = 0; i < a.length; i++)
7             for (int j = i+1; j < a.length; j++)
```

```
8         if (a[i] > a[j]) {
9             int temp = a[i];
10            a[i] = a[j];
11            a[j] = temp;
12        }
13    for (int i = 0; i < a.length; i++)
14        System.out.print(a[i] + " ");
15    }
16 }
```

Give the output of this program when running `java Q6 4 8 7 5 3 6`.

7. (3 points) How many bits are there in a **double** variable?

8. (6 points) What are the advantages of standard input stream compared to command-line arguments? (Name two advantages, one for 3 points. Reasonable answers are acceptable.)

Run $\text{\LaTeX}$ again to produce the table